Recommended Reading for Undergraduate Students in Physics and Mathematics

This list is organised first by language, then by subject. Nothing here is required reading: the intention is to point to books that are worth living with for a while, rather than to cover a syllabus. Titles link to the publisher's page where one exists, and otherwise to a bookshop search.

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1 Physics (Japanese)

1.1 History and Perspective

- 朝永振一郎『物理学とは何だろうか（上・下）』
  岩波新書 黄版 85・86、岩波書店、1979年 — How physics came to be: Kepler and Galileo, then Watt, Carnot and Clausius, ending in the kinetic theory of heat. Left unfinished at the author's death, and awarded the Ōsaragi Jirō Prize. Collected, together with his centenary lecture to the Physical Society of Japan, as 『朝永振一郎著作集 7』【新装版】（みすず書房、2001年）

1.2 Electromagnetism

- 砂川重信『電磁気学』
  物理テキストシリーズ 4、岩波書店、1987年 — A patient, physically motivated classic; the companion volume 『電磁気学演習』(同シリーズ 5) supplies worked problems

1.3 General Relativity

- 白水徹也『アインシュタイン方程式 — 一般相対性理論のよりよい理解のために』
  SGCライブラリ 90、サイエンス社、2012年 — A compact modern account, written with the AdS/CFT era in view

1.4 Particle Physics

- 南部陽一郎『クォーク — 素粒子物理はどこまで進んできたか』
  ブルーバックス、講談社（第2版、1998年） — Popular in form, but written by one of the architects of the subject

1.5 Thermodynamics and Statistical Mechanics

- 田崎晴明『熱力学 — 現代的な視点から』
  新物理学シリーズ 32、培風館、2000年 — An operational reconstruction resting on work and the Kelvin principle; the author's page carries errata and supplementary notes
- 清水明『熱力学の基礎［第2版］I・II』
  東京大学出版会、2021年（初版 2007年） — The same project begun from entropy rather than temperature, which buys a wider range of validity; volume II reaches phase transitions, chemical thermodynamics and systems in external fields
- 田崎晴明『統計力学 I・II』
  新物理学シリーズ 37・38、培風館、2008年 — Written to be read rather than consulted; volume I on its own already gives a coherent picture of the subject
- 清水明『統計力学の基礎 I』
  東京大学出版会、2024年（続巻準備中） — Takes typicality of equilibrium and Boltzmann's formula as the basic principles, treating ensembles as a calculational device rather than an axiom
- 高橋和孝『熱力学・統計力学 — 熱をめぐる諸相』
  講談社、2023年 — Treats thermodynamics and statistical mechanics as a single subject, with a generous supply of exercises

2 Mathematics (Japanese)

2.1 Geometry and Topology

- 長野正『曲面の数学 — 現代数学入門』
  培風館、1968年 — Surfaces approached from differential geometry, topology and complex analysis at once; an old book that has aged well
- 河野俊丈『曲面の幾何構造とモジュライ［増補版］』
  日本評論社、2023年（初版 1997年） — Hyperbolic structures, Teichmüller space and moduli; the enlarged edition adds a chapter on quantisation
- 服部晶夫『多様体（増補版）』
  岩波全書 288、岩波書店、1989年 — A concise first course on manifolds
- 服部晶夫『多様体のトポロジー［新装版］』
  岩波書店、2025年 — Morse theory, de Rham cohomology, degree and linking numbers, in under two hundred pages
- 森田茂之『微分形式の幾何学』
  岩波書店、2005年（岩波講座「現代数学の基礎」より単行本化） — From differential forms to characteristic classes, with the geometric meaning kept in the foreground

2.2 Algebra and Linear Algebra

- 堀田良之『加群十話 — 代数学入門』
  すうがくぶっくす 3、朝倉書店、1988年 — Ten informal chapters that reorganise algebra around the notion of a module, ending with $\mathcal{D}$-modules
- 久賀道郎『ガロアの夢 — 群論と微分方程式』
  日本評論社、1968年（ちくま学芸文庫より復刊） — Covering spaces, monodromy and Fuchsian equations, argued almost entirely in pictures
- 長谷川浩司『線型代数［改訂版］』
  日本評論社 — Goes well past the standard first course, towards representation theory and quadratic forms

3 Essays and Memoirs (Japanese)

- 藤原正彦『若き数学者のアメリカ』
  新潮文庫 — A young mathematician's first years abroad
- 藤原正彦『遥かなるケンブリッジ — 一数学者のイギリス』
  新潮文庫 — The Cambridge sequel, written a decade and a family later
- 小平邦彦『ボクは算数しか出来なかった』
  岩波現代文庫、2002年 — The autobiography of Japan's first Fields medallist, on how research can look from the inside
- 広中平祐『生きること学ぶこと』
  集英社文庫、1984年（新装版 2011年、解説＝小澤征爾・福岡伸一） — The Fields medallist of 1970 on curiosity, adversity and the pleasure of creating something; addressed to teenagers as much as to students

4 Physics (English)

4.1 General and Popular

- Richard P. Feynman, Robert B. Leighton, Matthew Sands, The Feynman Lectures on Physics, Vols. I–III.
  New Millennium edition, Basic Books, 2011; freely readable online — Best used alongside a conventional textbook rather than instead of one
- Richard P. Feynman, QED: The Strange Theory of Light and Matter.
  Princeton University Press, 1985 — Quantum electrodynamics with amplitudes drawn as little arrows, and no mathematics beyond addition
- Steven Weinberg, The First Three Minutes: A Modern View of the Origin of the Universe.
  2nd edition, Basic Books, 1993 — Still the model for how to write about cosmology honestly
- Stephen Hawking, A Brief History of Time.
  Bantam, 1988; updated edition 1998
- Kip S. Thorne, Black Holes and Time Warps: Einstein's Outrageous Legacy.
  Commonwealth Fund Book Program, W. W. Norton, 1994; foreword by Stephen Hawking, introduction by Frederick Seitz — The history of black-hole physics told from inside it; the preface maps out easy, intermediate and advanced routes through the book, so it can be read at several levels

4.2 Classical Mechanics

- L. D. Landau and E. M. Lifshitz, Mechanics.
  Course of Theoretical Physics, Vol. 1, 3rd edition, Butterworth–Heinemann, 1976 — Two hundred pages that derive most of mechanics from the action principle
- Herbert Goldstein, Charles Poole, John Safko, Classical Mechanics.
  3rd edition, Addison–Wesley, 2002 — The standard graduate-entrance text; strong on canonical transformations and rigid bodies
- V. I. Arnold, Mathematical Methods of Classical Mechanics.
  2nd edition, Graduate Texts in Mathematics 60, Springer, 1989 — Mechanics rebuilt on symplectic geometry; the natural bridge from physics to differential geometry

4.3 General Relativity and Classical Field Theory

- James B. Hartle, Gravity: An Introduction to Einstein's General Relativity.
  Addison–Wesley, 2003; reissued by Cambridge University Press, 2021 — The "physics-first" route into general relativity: black holes, gravitational waves and cosmology come before the Einstein equation, and mechanics is the only prerequisite
- L. D. Landau and E. M. Lifshitz, The Classical Theory of Fields.
  Course of Theoretical Physics, Vol. 2, 4th revised edition, Butterworth–Heinemann, 1975 — Special relativity, electrodynamics and general relativity in one continuous argument

4.4 Quantum Mechanics

- J. J. Sakurai and Jim Napolitano, Modern Quantum Mechanics.
  3rd edition, Cambridge University Press, 2020 — Symmetry-first, starting from spin rather than from wave mechanics

- J. J. Sakurai, Advanced Quantum Mechanics.
  Addison–Wesley, 1967 — An older route into relativistic quantum mechanics and field quantisation

4.5 Quantum Field Theory and Many-Body Physics

- Michael E. Peskin and Daniel V. Schroeder, An Introduction to Quantum Field Theory.
  Addison–Wesley, 1995 — The common language of the field; work the calculations rather than read them
- Piers Coleman, Introduction to Many-Body Physics.
  Cambridge University Press, 2015 — Field theory turned towards condensed matter: broken symmetry, superconductivity, the Kondo effect

5 Mathematics (English)

5.1 Visual and Intuitive Introductions

- Tristan Needham, Visual Complex Analysis.
  Oxford University Press, 1997; 25th Anniversary Edition, 2023 — Complex analysis done through geometry, with almost every theorem given a picture
- Tristan Needham, Visual Differential Geometry and Forms.
  Princeton University Press, 2021 — The same programme carried into curvature, holonomy and forms
- David Mumford, Caroline Series, David Wright, Indra's Pearls: The Vision of Felix Klein.
  Cambridge University Press, 2002 — Kleinian groups explored by computer, with the algorithms included
- Jeffrey R. Weeks, The Shape of Space.
  3rd edition, CRC Press, 2020 — Builds intuition for three-manifolds from scratch; no prerequisites

5.2 Geometry and Group Theory

- John Stillwell, The Four Pillars of Geometry.
  Undergraduate Texts in Mathematics, Springer, 2005 — Euclidean, linear-algebraic, projective and transformation-group views of the same subject
- John Stillwell, Geometry of Surfaces.
  Universitext, Springer, 1992 — The classification of surfaces via their geometries
- John Stillwell, Classical Topology and Combinatorial Group Theory.
  2nd edition, Graduate Texts in Mathematics 72, Springer, 1993 — Low-dimensional topology as it actually developed, knots and surfaces first
- John Stillwell, Naive Lie Theory.
  Undergraduate Texts in Mathematics, Springer, 2008 — Matrix groups and their Lie algebras with a minimum of machinery

5.3 Differential Topology and Characteristic Classes

- John Milnor, Topology from the Differentiable Viewpoint.

Princeton Landmarks in Mathematics, Princeton University Press, 1997 — Degree, Sard's theorem and the Poincaré–Hopf theorem in sixty pages

- John Milnor, Morse Theory.

  Annals of Mathematics Studies 51, Princeton University Press, 1963 — How a single function reveals the topology of a manifold; still the best account

- John Milnor and James D. Stasheff, Characteristic Classes.

  Annals of Mathematics Studies 76, Princeton University Press, 1974 — Stiefel–Whitney, Chern and Pontrjagin classes, done properly